

Introduction to Coordinates

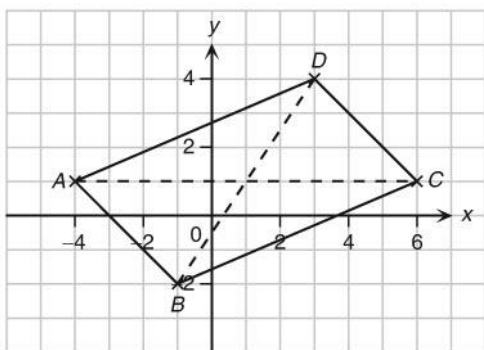
Teaching Examples

(1B10A001)

- (a) The coordinates of A are $(4, 3)$. A lies in quadrant I.
The coordinates of C are $(-3, -2)$. C lies in quadrant III.
- (b) x -coordinate of $B = \underline{\underline{-2}}$
- (c) y -coordinate of $D = \underline{\underline{-5}}$

(1B10A002)

(a), (b)



- (c) The coordinates of the point of intersection of the diagonals of $ABCD = \underline{\underline{(1, 1)}}$

(1B10A003)

$$PA = [1 - (-2)] \text{ units}$$

$$= (1 + 2) \text{ units}$$

$$= 3 \text{ units}$$

$$QA = (3 - 1) \text{ units}$$

$$= 2 \text{ units}$$

$$\therefore PA > QA$$

$\therefore$ Point P is further away from A .

(1B10A004)

$$AP = [-6.5 - (-16.5)] \text{ units}$$

$$= (-6.5 + 16.5) \text{ units}$$

$$= 10 \text{ units}$$

$$AQ = [6.5 - (-6.5)] \text{ units}$$

$$= (6.5 + 6.5) \text{ units}$$

$$= 13 \text{ units}$$

$$\therefore AQ > AP$$

$\therefore$ Point Q is further away from A .

(1B10A005)

$$PQ = [4 - (-1)] \text{ units} = 5 \text{ units}$$

$$QR = [9 - (-7)] \text{ units} = 16 \text{ units}$$

$$\begin{aligned} \therefore \text{ Perimeter of rectangle } PQRS &= 2(PQ + QR) \\ &= 2(5 + 16) \text{ units} \\ &= \underline{\underline{42 \text{ units}}} \end{aligned}$$

(1B10A006)

$$\begin{aligned} \text{(a)} \quad AB &= (10 - 4) \text{ units} \\ &= \underline{\underline{6 \text{ units}}} \end{aligned}$$

(b) $\because$ Points B , C and D are of the same distance from A .

$$\therefore AB = AC = AD$$

$$2 - c = 6$$

$$c = \underline{\underline{-4}}$$

(c) Let the coordinates of D be (a, b) .

$\because$ BAD is a vertical line.

$$\therefore a = 2$$

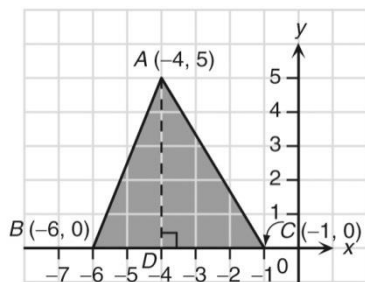
$$\because AD = 6 \text{ units}$$

$$\therefore 4 - b = 6$$

$$b = -2$$

$\therefore$ The coordinates of $D = \underline{\underline{(2, -2)}}$

(1B10A007)



Construct line segment AD such that $AD \perp BC$.

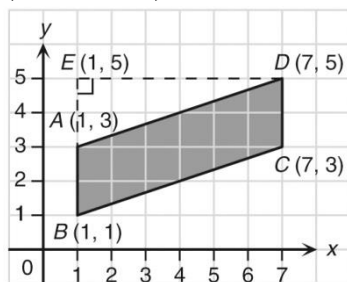
Then, the coordinates of $D = (-4, 0)$.

$$BC = [-1 - (-6)] \text{ units} = 5 \text{ units}$$

$$AD = (5 - 0) \text{ units} = 5 \text{ units}$$

$$\begin{aligned} \therefore \text{ Area of } \triangle ABC &= \frac{1}{2} \times BC \times AD \\ &= \frac{1}{2} \times 5 \times 5 \text{ sq. units} \\ &= \underline{\underline{12.5 \text{ sq. units}}} \end{aligned}$$

(1B10A008)



Construct a perpendicular line from D to meet BA produced at E .

Then, the coordinates of $E = (1, 5)$.

$$\therefore AB = (3 - 1) \text{ units} = 2 \text{ units}$$

$$DE = (7 - 1) \text{ units} = 6 \text{ units}$$

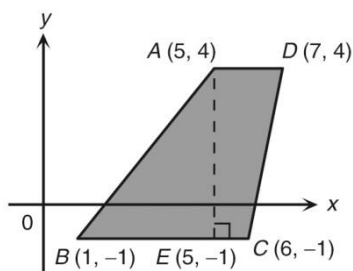
$$\therefore \text{Area of parallelogram } ABCD$$

$$= AB \times DE$$

$$= 2 \times 6 \text{ sq. units}$$

$$= \underline{\underline{12 \text{ sq. units}}}$$

(1B10A009)



Construct line segment AE such that $AE \perp BC$.

Then, the coordinates of $E = (5, -1)$.

$$AD = (7 - 5) \text{ units}$$

$$= 2 \text{ units}$$

$$BC = (6 - 1) \text{ units}$$

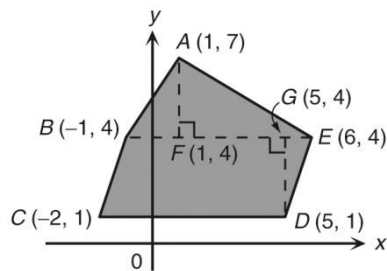
$$= 5 \text{ units}$$

$$AE = [4 - (-1)] \text{ units}$$

$$= 5 \text{ units}$$

$$\begin{aligned} \therefore \text{Area of trapezium } ABCD &= \frac{1}{2} \times (AD + BC) \times AE \\ &= \frac{1}{2} \times (2 + 5) \times 5 \text{ sq. units} \\ &= \underline{\underline{17.5 \text{ sq. units}}} \end{aligned}$$

(1B10A010)



Take BE as the common base of $\triangle ABE$ and parallelogram $BCDE$. Draw the height AF of $\triangle ABE$ and the height DG of parallelogram $BCDE$.

Then, the coordinates of $F = (1, 4)$,

the coordinates of $G = (5, 4)$.

$$BE = [6 - (-1)] \text{ units}$$

$$= 7 \text{ units}$$

$$AF = (7 - 4) \text{ units}$$

$$= 3 \text{ units}$$

$$DG = (4 - 1) \text{ units}$$

$$= 3 \text{ units}$$

$\therefore$ Area of $ABCDE$ = area of $\triangle ABE$ + area of parallelogram $BCDE$

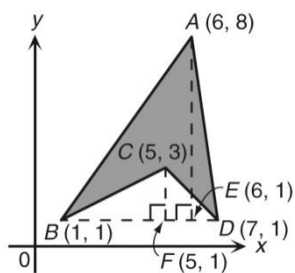
$$= \frac{1}{2} \times BE \times AF + BE \times DG$$

$$= \left(\frac{1}{2} \times 7 \times 3 + 7 \times 3 \right) \text{ sq. units}$$

$$= (10.5 + 21) \text{ sq. units}$$

$$= \underline{\underline{31.5 \text{ sq. units}}}$$

(1B10A011)



Join BD . Construct line segments AE and CF such that $AE \perp BD$ and $CF \perp BD$ respectively.

Then, we have $E(6, 1)$ and $F(5, 1)$.

$$\text{Area of } \triangle ABD = \frac{1}{2} \times BD \times AE$$

$$= \frac{1}{2} \times 6 \times 7 \text{ sq. units}$$

$$= 21 \text{ sq. units}$$

$$\begin{aligned}
 \text{Area of } \triangle CBD &= \frac{1}{2} \times BD \times CF \\
 &= \frac{1}{2} \times 6 \times 2 \text{ sq. units} \\
 &= 6 \text{ sq. units}
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{Area of } ABCD &= \text{area of } \triangle ABD - \text{area of } \triangle CBD \\
 &= (21 - 6) \text{ sq. units} \\
 &= \underline{\underline{15 \text{ sq. units}}}
 \end{aligned}$$

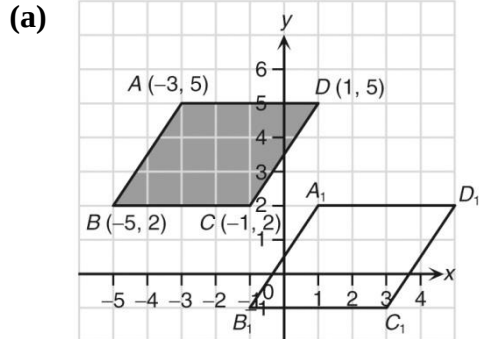
(1B10A012)

(a) The coordinates of $P_1 = (5 + 3, 8)$
 $= \underline{\underline{(8, 8)}}$

(b) The coordinates of $P_2 = (8, 8 - 1)$
 $= \underline{\underline{(8, 7)}}$

(c) The coordinates of $P_3 = (8 - 6, 7)$
 $= \underline{\underline{(2, 7)}}$

(1B10A013)



(b) From the figure, we find that:

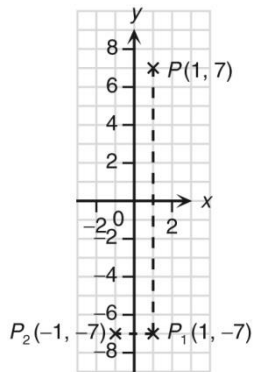
The coordinates of $A_1 = \underline{\underline{(1, 2)}}$

The coordinates of $B_1 = \underline{\underline{(-1, -1)}}$

The coordinates of $C_1 = \underline{\underline{(3, -1)}}$

The coordinates of $D_1 = \underline{\underline{(5, 2)}}$

(1B10A014)



The coordinates of $P_1 = \underline{\underline{(1, -7)}}$

The coordinates of $P_2 = \underline{\underline{(-1, -7)}}$

(1B10A015)

The distance between P' and line L
 = the distance between point P and line L
 = $(10 - 3)$ units
 = 7 units

The coordinates of $P' = (7, 10 - 2 \times 7)$
 = $\underline{\underline{(7, -4)}}$

(1B10A016)

The coordinates of $A' = (-(-1), -6)$
 = $\underline{\underline{(1, -6)}}$

(1B10A017)

(a) The coordinates of $B = (-4, -(-9))$
 = $\underline{\underline{(-4, 9)}}$

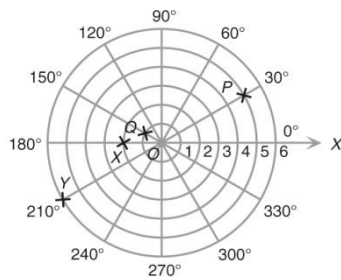
(b) The coordinates of $C = (9, -(-4))$
 = $\underline{\underline{(9, 4)}}$

(1B10A018)

(a) The required polar coordinates are:

$$P(5, 30^\circ), Q(1, 150^\circ)$$

(b)

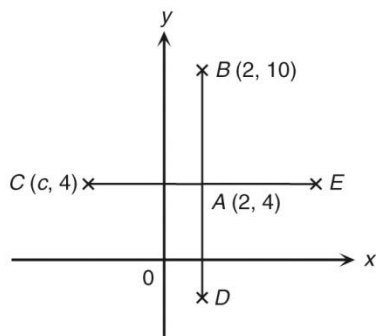


$$\begin{aligned} \text{(c)} \quad \angle XOY &= 210^\circ - 180^\circ \\ &= 30^\circ \end{aligned}$$

$$PY = (5 + 6) \text{ units}$$

$$= \underline{\underline{11 \text{ units}}}$$

(1B10A019)

Point E may lie on the right of A as shown in the figure.Let the coordinates of E be (e, f) . $\therefore CAE$ is a horizontal line.

$$\therefore f = 4$$

 $\therefore$ Point E is of the same distance from A as points B , C and D .

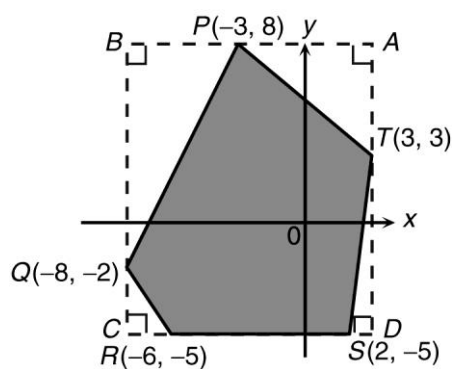
$$\therefore AE = 6 \text{ units}$$

$$e - 2 = 6$$

$$e = 8$$

 $\therefore$ The coordinates of $E = \underline{\underline{(8, 4)}}$ (or any other reasonable answers)

(1B10A020)



Construct a rectangle $ABCD$ such that P , Q and T lie on AB , BC and DA respectively.

Then, we have $A(3, 8)$, $B(-8, 8)$, $C(-8, -5)$ and $D(3, -5)$.

$$\begin{aligned}
 \text{Area of rectangle } ABCD &= AB \times BC \\
 &= 11 \times 13 \text{ sq. units} \\
 &= 143 \text{ sq. units}
 \end{aligned}$$

$$\begin{aligned}
 \text{Area of } \triangle APT &= \frac{1}{2} \times AP \times AT \\
 &= \frac{1}{2} \times 6 \times 5 \text{ sq. units} \\
 &= 15 \text{ sq. units}
 \end{aligned}$$

$$\begin{aligned}
 \text{Area of } \triangle BQP &= \frac{1}{2} \times BP \times BQ \\
 &= \frac{1}{2} \times 5 \times 10 \text{ sq. units} \\
 &= 25 \text{ sq. units}
 \end{aligned}$$

$$\begin{aligned}
 \text{Area of } \triangle CRQ &= \frac{1}{2} \times CQ \times CR \\
 &= \frac{1}{2} \times 3 \times 2 \text{ sq. units} \\
 &= 3 \text{ sq. units}
 \end{aligned}$$

$$\begin{aligned}
 \text{Area of } \triangle DTS &= \frac{1}{2} \times DS \times DT \\
 &= \frac{1}{2} \times 1 \times 8 \text{ sq. units} \\
 &= 4 \text{ sq. units}
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{Area of pentagon } PQRST & \\
 &= [143 - (15 + 25 + 3 + 4)] \text{ sq. units} \\
 &= \underline{\underline{96 \text{ sq. units}}}
 \end{aligned}$$

(1B10A021)

The coordinates of $P' = \underline{\underline{(-9, -2)}}$

Level 2 Questions

(1B10D001)

(a) $\because$ A lies on the x -axis. $\therefore$ y -coordinate of $A = 0$ $\therefore m - 5 = 0$

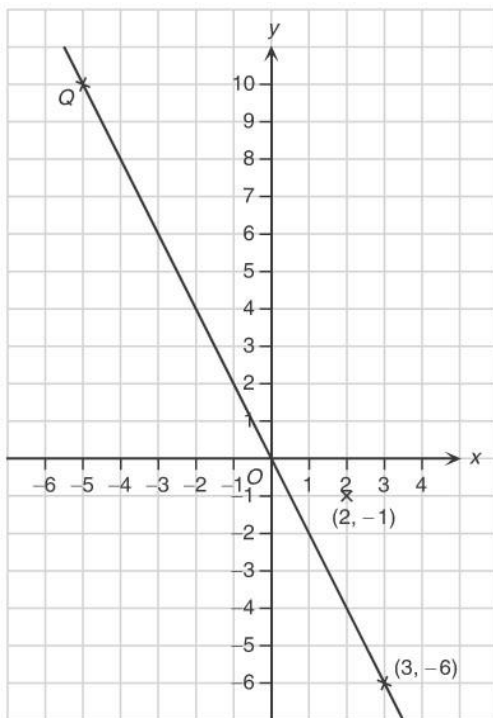
$$m = 5$$

 $\therefore$ The coordinates of $A = (2(5) + 1, (5) - 5)$
 $= \underline{\underline{(11, 0)}}$ **(b)** $\because$ A lies on the y -axis. $\therefore$ x -coordinate of $A = 0$ $\therefore 2m + 1 = 0$

$$m = -\frac{1}{2}$$

 $\therefore$ The coordinates of $A = \left(2\left(-\frac{1}{2}\right) + 1, \left(-\frac{1}{2} - 5\right) \right)$
 $= \underline{\underline{\left(0, -5\frac{1}{2} \right)}}$

(1B10D002)

(a)**(b) (i)** No**(ii)** Yes

(1B10D003)

(a) Coordinates of $B = \underline{\underline{(-3, -2)}}$ (b) P lies on the side of $\triangle ABC$. Q lies inside $\triangle ABC$. R lies outside $\triangle ABC$.

(1B10D004)

(a) $AB = [3 - (-2)]$ units = 5 units $BC = [4 - (-3)]$ units = 7 units $CD = [3 - (-4)]$ units = 7 units $DE = [1 - (-3)]$ units = 4 units(b) Line segment DE is the shortest.

(1B10D005)

(a) The coordinates of $B = \underline{\underline{(3, -2)}}$ (b) $AB = [3 - (-6)]$ units = 9 units $BC = [4 - (-2)]$ units = 6 units

(1B10D006)

(a) $AB = [2 - (-5)]$ units = 7 units $BC = [6 - (-1)]$ units = 7 units $CD = [2 - (-5)]$ units = 7 units $DA = [6 - (-1)]$ units = 7 units(b) $ABCD$ is a square.

(c) Perimeter of square $ABCD = AB \times 4$
 $= (7 \times 4)$ units
 $= \underline{\underline{28 \text{ units}}}$

(1B10D007)

(a) $PQ = [-1.5 - (-5.5)]$ units = 4 units $QR = [6.5 - (-3)]$ units = 9.5 units $RS = [-1.5 - (-5.5)]$ units = 4 units $SP = [6.5 - (-3)]$ units = 9.5 units(b) $PQRS$ is a rectangle.

(c) Perimeter of quadrilateral $PQRS = (PQ + QR) \times 2$
 $= (4 + 9.5) \times 2$ units
 $= \underline{\underline{27 \text{ units}}}$

(1B10D008)

(a) The coordinates of $P = \underline{\underline{(5, -4)}}$,the coordinates of $R = \underline{\underline{(11, -1)}}$,the coordinates of $S = \underline{\underline{(5, -1)}}$ **(b)** $PQ = (11 - 5) \text{ units} = 6 \text{ units}$ $SP = [-1 - (-4)] \text{ units} = 3 \text{ units}$

$$\begin{aligned} \text{Perimeter of } PQRS &= (PQ + SP) \times 2 \\ &= (6 + 3) \times 2 \text{ units} \\ &= \underline{\underline{18 \text{ units}}} \end{aligned}$$

(1B10D009)

(a) The coordinates of $B = \underline{\underline{(1, -2)}}$,The coordinates of $D = \underline{\underline{(6, 3)}}$ **(b)** $AB = [3 - (-2)] \text{ units} = 5 \text{ units}$ $BC = (6 - 1) \text{ units} = 5 \text{ units}$

$$\begin{aligned} \text{Perimeter of rectangle } ABCD &= (AB + BC) \times 2 \\ &= (5 + 5) \times 2 \text{ units} \\ &= \underline{\underline{20 \text{ units}}} \end{aligned}$$

(1B10D010)

 $\therefore AB = 9 \text{ units}$

$$\therefore 3a + 1 - (-2) = 9$$

$$3a + 3 = 9$$

$$3a = 6$$

$$a = \underline{\underline{2}}$$

(1B10D011)

 $\therefore CD = 14 \text{ units}$

$$\therefore 7 - (-2d + 3) = 14$$

$$2d - 3 = 7$$

$$2d = 10$$

$$d = \underline{\underline{5}}$$

(1B10D012)

$$\because AB = 18 \text{ units}$$

$$\therefore 4h + 5 - (2 - h) = 18$$

$$4h + 5 - 2 + h = 18$$

$$5h = 15$$

$$h = \underline{\underline{3}}$$

(1B10D013)

$$\because CD = 8 \text{ units}$$

$$\therefore h - 3 - (-2h + 1) = 8$$

$$h - 3 + 2h - 1 = 8$$

$$3h = 12$$

$$h = \underline{\underline{4}}$$

(1B10D014)

$$\because AB = 9 \text{ units}$$

$$\therefore 4 - a = 9$$

$$-a = 9 - 4$$

$$a = \underline{\underline{-5}}$$

$$\because BC = 8 \text{ units}$$

$$\therefore c - (-3) = 8$$

$$c + 3 = 8$$

$$c = \underline{\underline{5}}$$

(1B10D015)

$$\because AB = 7 \text{ units}$$

$$\therefore 4 - (-4x + 1) = 7$$

$$4 + 4x - 1 = 7$$

$$4x + 3 = 7$$

$$4x = 4$$

$$x = \underline{\underline{1}}$$

$$\because BC = 5 \text{ units}$$

$$\therefore 3 - 2y = 5$$

$$-2y = 5 - 3$$

$$y = \underline{\underline{-1}}$$

(1B10D016)

 $\therefore$ Perimeter of square $PQRS = 16$ units $\therefore PQ = QR = RS = SP = 4$ unitsThe coordinates of $Q = (-5, 2 - 4) = \underline{\underline{(-5, -2)}}$ The coordinates of $R = (4 + (-5), -2) = \underline{\underline{(-1, -2)}}$ The coordinates of $S = (x\text{-coordinate of } R, y\text{-coordinate of } P) = \underline{\underline{(-1, 2)}}$

(1B10D017)

$$\begin{aligned}
 \text{(a) } AB &= [(6 + 2x) - 2x] \text{ unit} \\
 &= (6 + 2x - 2x) \text{ units} \\
 &= \underline{\underline{6 \text{ units}}}
 \end{aligned}$$

$$\text{(b) } \therefore CD = AB = 6 \text{ units}$$

$$\begin{aligned}
 \therefore [(2y - 1) - (-1)] &= 6 \\
 2y &= 6 \\
 y &= \underline{\underline{3}}
 \end{aligned}$$

$$\begin{aligned}
 \therefore x\text{-coordinate of } B &= x\text{-coordinate of } C \\
 \therefore 6 + 2x &= 2y - 1 \\
 6 + 2x &= 2(3) - 1 \\
 2x &= -1 \\
 x &= -\underline{\underline{\frac{1}{2}}}
 \end{aligned}$$

(1B10D018)

$$AB + CD + EF = [6 - (-5)] \text{ units} = 11 \text{ units}$$

$$JC + IE + HG = [6 - (-5)] \text{ units} = 11 \text{ units}$$

$$JA + IC + HE = [5 - (-3)] \text{ units} = 8 \text{ units}$$

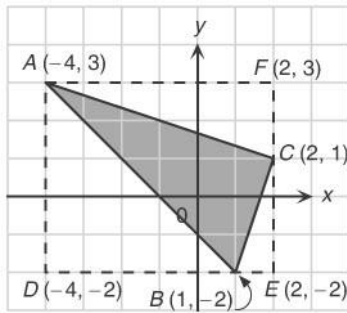
$$CB + ED + GF = [5 - (-3)] \text{ units} = 8 \text{ units}$$

Perimeter of the figure

$$= (11 + 11 + 8 + 8) \text{ units}$$

$$= \underline{\underline{38 \text{ units}}}$$

(1B10D019)



Divide the figure into $\triangle ACD$ and $\triangle BCD$.

Then, the coordinates of $D = (-2, 1)$

For $\triangle ACD$,

$$CD = [2 - (-2)] \text{ units} = 4 \text{ units}$$

$$\begin{aligned} \text{height of } \triangle ACD &= (3 - 1) \text{ units} \\ &= 2 \text{ units} \end{aligned}$$

$$\begin{aligned} \text{area of } \triangle ACD &= \frac{1}{2} \times 4 \times 2 \text{ sq. units} \\ &= 4 \text{ sq. units} \end{aligned}$$

For $\triangle BCD$,

$$\begin{aligned} \text{height of } \triangle BCD &= [1 - (-2)] \text{ units} \\ &= 3 \text{ units} \end{aligned}$$

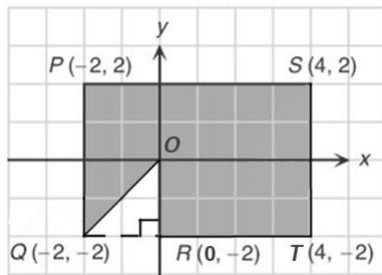
$$\begin{aligned} \text{area of } \triangle BCD &= \frac{1}{2} \times 4 \times 3 \text{ sq. units} \\ &= 6 \text{ sq. units} \end{aligned}$$

$$\therefore \text{Area of } \triangle ABC = \text{area of } \triangle ACD + \text{area of } \triangle BCD$$

$$= (4 + 6) \text{ sq. units}$$

$$= \underline{\underline{10 \text{ sq. units}}}$$

(1B10D020)



Construct a line QR which is perpendicular to OR .

$$PQ = [2 - (-2)] \text{ units} = 4 \text{ units}$$

$$PS = [4 - (-2)] \text{ units} = 6 \text{ units}$$

$$QR = [0 - (-2)] \text{ units} = 2 \text{ units}$$

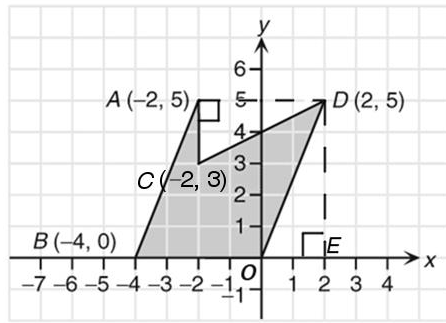
$$RO = [0 - (-2)] \text{ units} = 2 \text{ units}$$

$$\begin{aligned} \text{Area of rectangle } PQTS &= PQ \times PS \\ &= 4 \times 6 \text{ sq. units} \\ &= 24 \text{ sq. units} \end{aligned}$$

$$\begin{aligned} \text{Area of } \triangle OQR &= \frac{1}{2} \times QR \times RO \\ &= \frac{1}{2} \times 2 \times 2 \text{ sq. units} \\ &= 2 \text{ sq. units} \end{aligned}$$

$$\begin{aligned} \therefore \text{Area of the figure} &= \text{area of rectangle } PQTS - \text{area of } \triangle OQR \\ &= 24 - 2 \text{ sq. units} \\ &= \underline{\underline{22 \text{ sq. units}}} \end{aligned}$$

(1B10D021)



Construct a line AD which is perpendicular to AC and a line DE which is perpendicular to OB .

$$OB = [0 - (-4)] \text{ units} = 4 \text{ units}$$

$$DE = (5 - 0) \text{ units} = 5 \text{ units}$$

$$AC = (5 - 3) \text{ units} = 2 \text{ units}$$

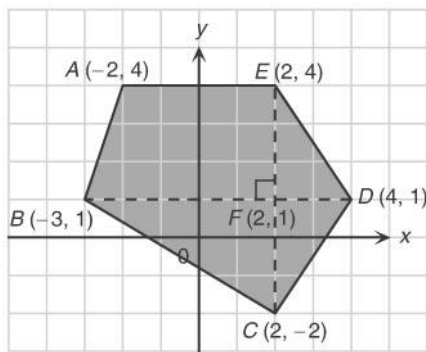
$$AD = [2 - (-2)] \text{ units} = 4 \text{ units}$$

$$\begin{aligned} \text{Area of parallelogram } ABOD &= OB \times DE \\ &= 4 \times 5 \text{ sq. units} \\ &= 20 \text{ sq. units} \end{aligned}$$

$$\begin{aligned} \text{Area of } \triangle ACD &= \frac{1}{2} \times AC \times AD \\ &= \frac{1}{2} \times 2 \times 4 \text{ sq. units} \\ &= 4 \text{ sq. units} \end{aligned}$$

$$\begin{aligned} \text{Area of the figure} &= \text{area of parallelogram } ABOD - \text{area of } \triangle ACD \\ &= 20 - 4 \text{ sq. units} \\ &= \underline{\underline{16 \text{ sq. units}}} \end{aligned}$$

(1B10D022)



Take BD as the common base of trapezium $ABDE$ and $\triangle BCD$. Draw the height EF of trapezium $ABDE$ and the height of CF of $\triangle BCD$.

Then, the coordinates of $F = (2, 1)$.

$$AE = [2 - (-2)] \text{ units} = 4 \text{ units}$$

$$BD = [4 - (-3)] \text{ units} = 7 \text{ units}$$

$$EF = (4 - 1) \text{ units} = 3 \text{ units}$$

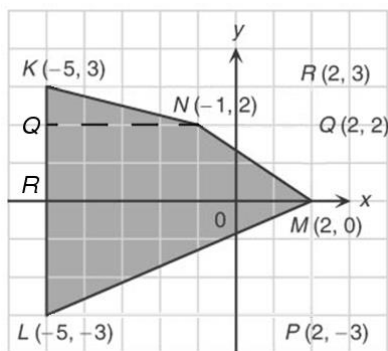
$$CF = [1 - (-2)] \text{ units} = 3 \text{ units}$$

$$\begin{aligned} \text{Area of trapezium } ABDE &= \frac{1}{2} \times (AE + BD) \times EF \\ &= \frac{1}{2} \times (4 + 7) \times 3 \text{ sq. units} \\ &= 16.5 \text{ sq. units} \end{aligned}$$

$$\begin{aligned} \text{Area of } \triangle BCD &= \frac{1}{2} \times BD \times CF \\ &= \frac{1}{2} \times 7 \times 3 \text{ sq. units} \\ &= 10.5 \text{ sq. units} \end{aligned}$$

$$\begin{aligned} \therefore \text{Area of pentagon } ABCDE &= \text{area of trapezium } ABDE + \text{area of } \triangle BCD \\ &= (16.5 + 10.5) \text{ sq. units} \\ &= \underline{\underline{27 \text{ sq. units}}} \end{aligned}$$

(1B10D023)



Construct line segment NQ such that $NQ \perp KL$ and let the point of intersection of KL and x -axis be R .

Then, we have $Q(-5, 2)$ and $R(-5, 0)$.

$$NQ = [-1 - (-5)] \text{ units}$$

$$= 4 \text{ units}$$

$$MR = [2 - (-5)] \text{ units}$$

$$= 7 \text{ units}$$

$$KQ = (3 - 2) \text{ unit}$$

$$= 1 \text{ unit}$$

$$QR = (2 - 0) \text{ units}$$

$$= 2 \text{ units}$$

$$RL = [0 - (-3)] \text{ units}$$

$$= 3 \text{ units}$$

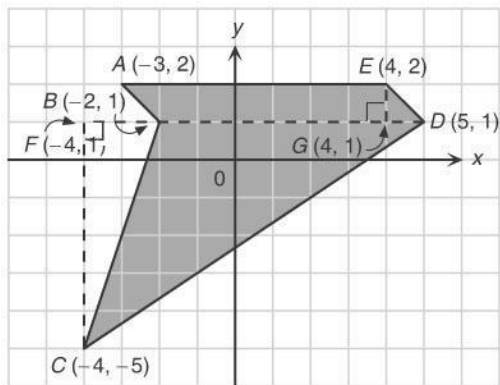
$$\begin{aligned} \text{Area of } \triangle KQN &= \frac{1}{2} \times NQ \times KQ \\ &= \frac{1}{2} \times 4 \times 1 \text{ sq. units} \\ &= 2 \text{ sq. units} \end{aligned}$$

$$\begin{aligned} \text{Area of trapezium } NQRM &= \frac{1}{2} \times (NQ + MR) \times QR \\ &= \frac{1}{2} \times (4 + 7) \times 2 \text{ sq. units} \\ &= 11 \text{ sq. units} \end{aligned}$$

$$\begin{aligned} \text{Area of } \triangle MRL &= \frac{1}{2} \times MR \times RL \\ &= \frac{1}{2} \times 7 \times 3 \text{ sq. units} \\ &= 10.5 \text{ sq. units} \end{aligned}$$

$$\begin{aligned} \therefore \text{Area of quadrilateral } KLMN &= \text{area of } \triangle KQN + \text{area of trapezium } NQRM + \text{area of } \triangle MRL \\ &= (2 + 11 + 10.5) \text{ sq. units} \\ &= \underline{\underline{23.5 \text{ sq. units}}} \end{aligned}$$

(1B10D024)



Take BD as the common base of parallelogram $ABDE$ and $\triangle BCD$. Draw the height EG of parallelogram $ABDE$ and the height of CF of $\triangle BCD$.

Then, we have $F(-4, 1)$ and $G(4, 1)$.

$$BD = [5 - (-2)] \text{ units} = 7 \text{ units}$$

$$EG = (2 - 1) \text{ unit} = 1 \text{ unit}$$

$$CF = [5 - (-1)] \text{ units} = 6 \text{ units}$$

$$\text{Area of parallelogram } ABDE = BD \times EG$$

$$= 7 \times 1 \text{ sq. units}$$

$$= 7 \text{ sq. units}$$

$$\text{Area of } \triangle BCD = \frac{1}{2} \times BD \times CF$$

$$= \frac{1}{2} \times 7 \times 6 \text{ sq. units}$$

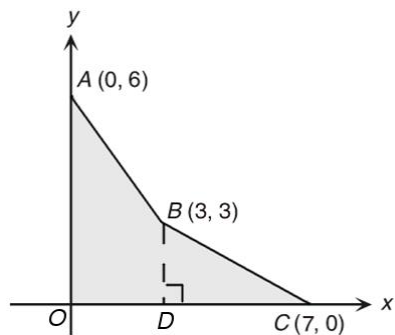
$$= 21 \text{ sq. units}$$

$$\therefore \text{Area of pentagon } ABCDE = \text{area of parallelogram } ABDE + \text{area of } \triangle BCD$$

$$= (7 + 21) \text{ sq. units}$$

$$= \underline{\underline{28 \text{ sq. units}}}$$

(1B10D025)



Construct a perpendicular line from B to meet OC produced at D .

Then the coordinates of $D = (3, 0)$.

$$AO = (6 - 0) \text{ units} = 6 \text{ units}$$

$$DO = (3 - 0) \text{ units} = 3 \text{ units}$$

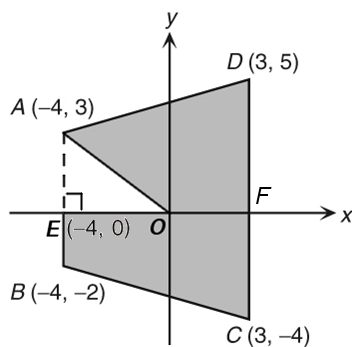
$$BD = (3 - 0) \text{ units} = 3 \text{ units}$$

$$CD = (7 - 3) \text{ units} = 4 \text{ units}$$

Area of the quadrilateral $OABC$ = area of $ODBA$ + area of $\triangle BDC$

$$\begin{aligned} &= \frac{1}{2} \times (BD + AO) \times DO + \frac{1}{2} \times BD \times CD \\ &= \left[\frac{1}{2} \times (3 + 6) \times 3 + \frac{1}{2} \times 3 \times 4 \right] \text{sq. units} \\ &= (13.5 + 6) \text{sq. units} \\ &= \underline{\underline{19.5 \text{sq. units}}} \end{aligned}$$

(1B10D026)



Construct a perpendicular line AE to meet OE .

Mark the point of intersection of CD and x -axis be F .

$$AB = [3 - (-2)] \text{ units} = 5 \text{ units}$$

$$AE = (3 - 0) \text{ units} = 3 \text{ units}$$

$$OE = [0 - (-4)] \text{ units} = 4 \text{ units}$$

$$DC = [5 - (-4)] \text{ units} = 9 \text{ units}$$

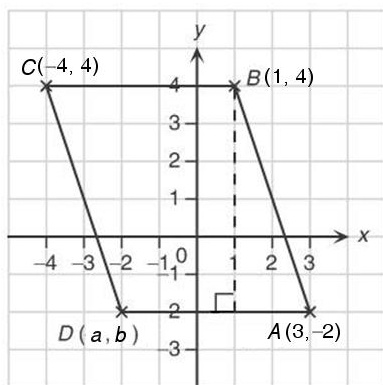
$$EF = [3 - (-4)] \text{ units} = 7 \text{ units}$$

Area of the figure = area of trapezium $ABCD$ – area of $\triangle AEO$

$$\begin{aligned}
 &= \frac{1}{2} \times (AB + DC) \times EF - \frac{1}{2} \times AE \times OE \\
 &= \left[\frac{1}{2} \times (5 + 9) \times 7 - \frac{1}{2} \times 3 \times 4 \right] \text{sq. units} \\
 &= 49 - 6 \text{sq. units} \\
 &= \underline{\underline{43 \text{sq. units}}}
 \end{aligned}$$

(1B10D027)

(a)



$$BC = [1 - (-4)] \text{ units} = 5 \text{ units}$$

Let the coordinates of D be (a, b) .

$$\therefore AD \parallel BC$$

$$\therefore \text{y-coordinate of } D = \text{y-coordinate of } A$$

$$\therefore b = -2$$

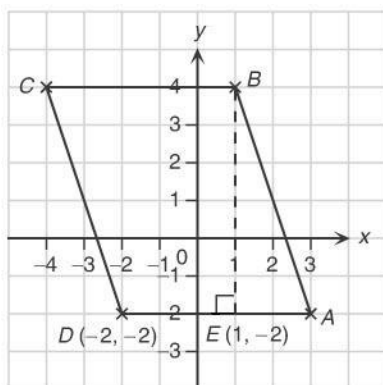
$$\therefore AD = BC \text{ and } D \text{ lies on the left of } A.$$

$$\therefore 3 - a = 5$$

$$a = -2$$

$$\therefore \text{Coordinates of } D = \underline{\underline{(-2, -2)}}$$

(b)



Construct line segment BE such that $BE \perp AD$.

Then, the coordinates of $E = (1, -2)$.

$$AD = BD = 5 \text{ units}$$

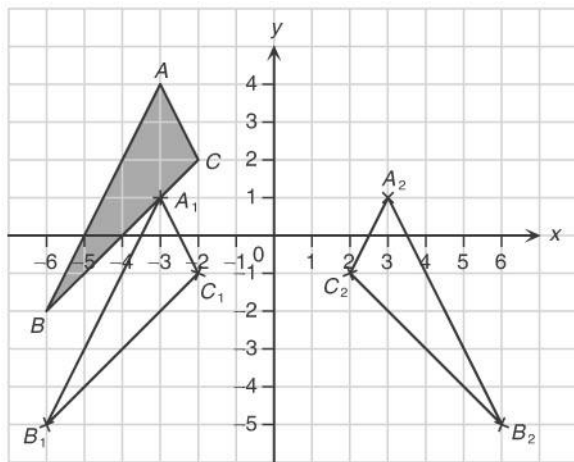
$$BE = [4 - (-2)] \text{ units} = 6 \text{ units}$$

$$\text{Area of parallelogram } ABCD = AD \times BE$$

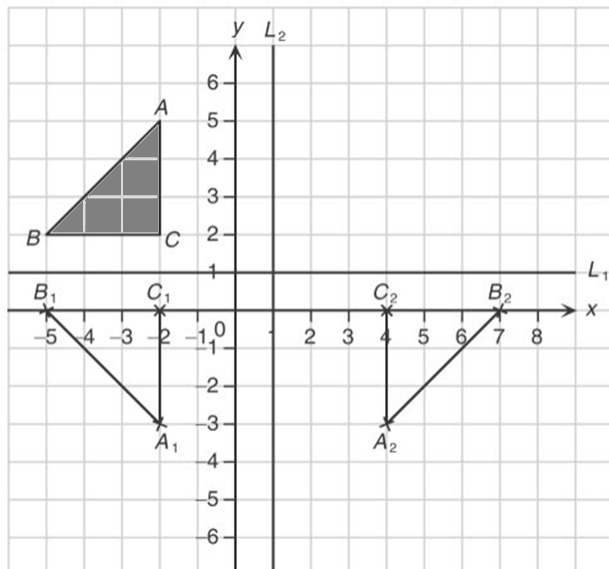
$$= 5 \times 6 \text{sq. units}$$

$$= \underline{\underline{30 \text{sq. units}}}$$

(1B10D028)

(a) $A_1(-3, 1)$, $B_1(-6, -5)$, $C_1(-2, -1)$ (b) $A_2(3, 1)$, $B_2(6, -5)$, $C_2(2, -1)$

(1B10D029)

(a) $A_1(-2, -3)$, $B_1(-5, 0)$, $C_1(-2, 0)$ (b) $A_2(4, -3)$, $B_2(7, 0)$, $C_2(4, 0)$

(1B10D030)

(a) $\because$ A is reflected about the x-axis to B.

$$\therefore x + 2 = -2$$

$$x = \underline{\underline{-4}}$$

$$\therefore -(y - 1) = 3$$

$$y = \underline{\underline{-2}}$$

(b) $\therefore$ The coordinates of A = $(-4 + 2, 3) = \underline{\underline{(-2, 3)}}$,
the coordinates of B = $(-2, -2-1) = \underline{\underline{(-2, -3)}}$

(1B10D031)

(a) $\therefore$ P is reflected about the y -axis to Q .

$$\therefore -x + 2 = -3$$

$$-x = -5$$

$$x = \underline{\underline{5}}$$

$$\therefore 2y + 1 = 5$$

$$2y = 4$$

$$y = \underline{\underline{2}}$$

(b) $\therefore$ The coordinates of $P = (3, 2(2) + 1) = \underline{\underline{(3, 5)}}$,
the coordinates of $Q = (-5 + 2, 5) = \underline{\underline{(-3, 5)}}$

(1B10D032)

(a) $\therefore$ M is translated to the left by 5 units to N .

$$\therefore 1 - 5 = 3 + x$$

$$-4 = 3 + x$$

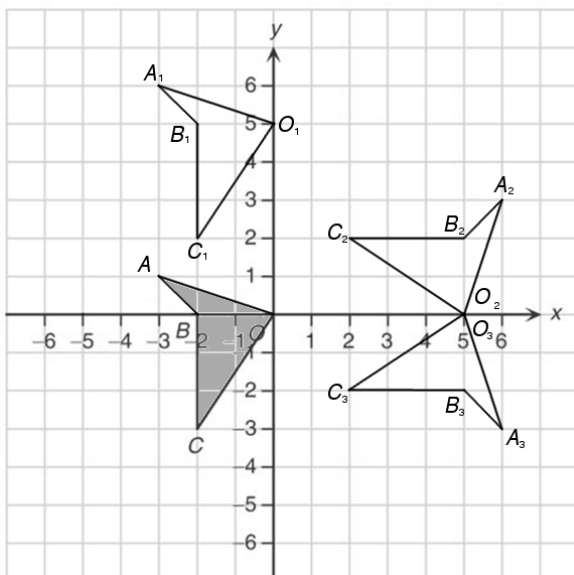
$$x = \underline{\underline{-7}}$$

$$\therefore 2 - y = 5$$

$$y = \underline{\underline{-3}}$$

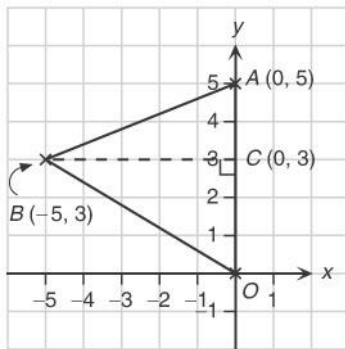
(b) $\therefore$ The coordinates of $M = (1, 2 - (-3)) = \underline{\underline{(1, 5)}}$,
the coordinates of $N = (3 + (-7), 5) = \underline{\underline{(-4, 5)}}$

(1B10D033)



(a) $O_1(0, 5), A_1(-3, 6), B_1(-2, 5), C_1(-2, 2)$ (b) $O_2(5, 0), A_2(6, 3), B_2(5, 2), C_2(2, 2)$ (c) $O_3(5, 0), A_3(6, -3), B_3(5, -2), C_3(2, -2)$

(1B10D034)

(a) When $A(0, 5)$ is rotated anti-clockwise about the origin through 90° , the coordinates of the image are $(-5, 0)$. Then $(-5, 0)$ is translated upwards by 3 units to B .The coordinates of $B = (-5, 0 + 3) = \underline{\underline{(-5, 3)}}$.(b) Construct line segment BC such that $BC \perp OA$.Then, the coordinates of $C = (0, 3)$

$$\begin{aligned}
 \therefore \text{Area of } \triangle OAB &= \frac{1}{2} \times OA \times BC \\
 &= \frac{1}{2} \times (5 - 0) \times [0 - (-5)] \text{ sq. units} \\
 &= \frac{1}{2} \times 5 \times 5 \text{ sq. units} \\
 &= \underline{\underline{12.5 \text{ sq. units}}}
 \end{aligned}$$

(1B10D035)

(a) $\therefore M$ is rotated anti-clockwise about the origin through 180° to N .

$$\therefore -(-x + 2) = 1$$

$$x - 2 = 1$$

$$x = \underline{\underline{3}}$$

$$\therefore -(3y + 1) = 2$$

$$-3y = 3$$

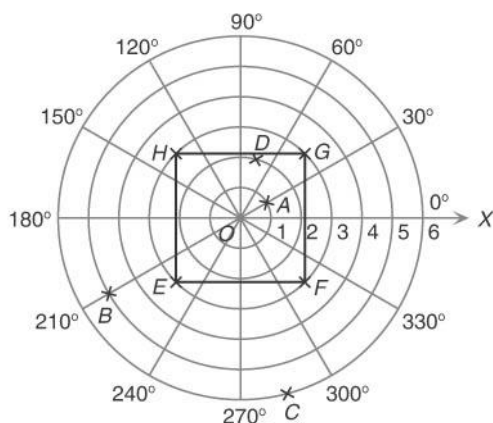
$$y = \underline{\underline{-1}}$$

(b) $\therefore$ The coordinates of $M = (1, 3(-1) + 1) = \underline{\underline{(1, -2)}}$,
the coordinates of $N = (-3 + 2, 2) = \underline{\underline{(-1, 2)}}$

(1B10D036)

(a) $A(1, 30^\circ)$, $B(5, 210^\circ)$, $C(6, 285^\circ)$, $D(2, 75^\circ)$

(b)

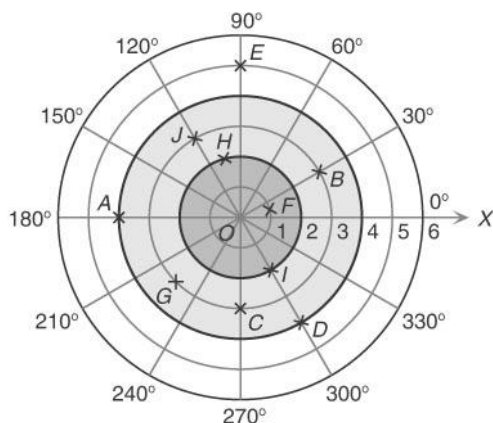


(c) $G(3, 45^\circ)$, $H(3, 135^\circ)$

(1B10D037)

(a) $A(4, 180^\circ)$, $B(3, 30^\circ)$, $C(3, 270^\circ)$, $D(4, 300^\circ)$, $E(5, 90^\circ)$

(b)



(c) Total distance of Helen's arrows from $O = (4 + 3 + 3 + 4 + 5)$ units = 19 units

Total distance of Eva's arrows from $O = (1 + 3 + 2 + 2 + 3)$ units = 11 units

$\therefore$ Eva is the winner.

(1B10D038)

(a) $\because \angle POQ = 90^\circ$

$\therefore$ The possible polar coordinates of Q are $(5, 60^\circ + 90^\circ)$ and $(5, 360^\circ + 60^\circ - 90^\circ)$,
i.e. $(5, 150^\circ)$ and $(5, 330^\circ)$.

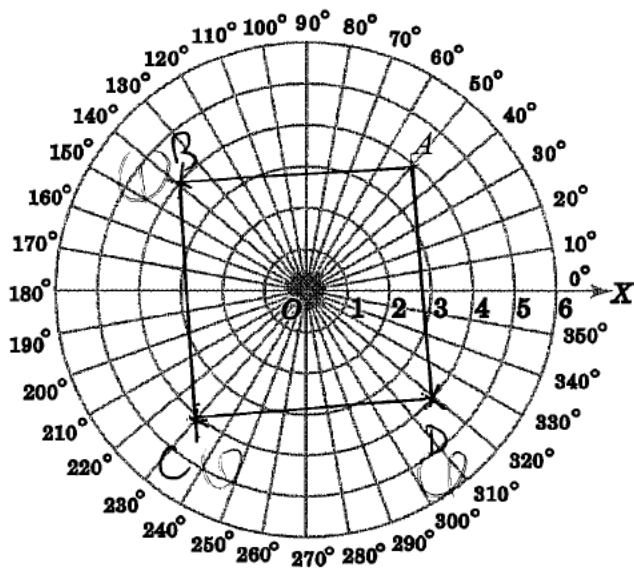
(b) Area of $\triangle POQ = \frac{1}{2} \times OP \times OQ$
 $= \frac{1}{2} \times 4 \times 5$ sq. units
 $= 10$ sq. units

Level 3 Questions

(1B10E001)

(a) $A = (4, 50^\circ)$

(b)

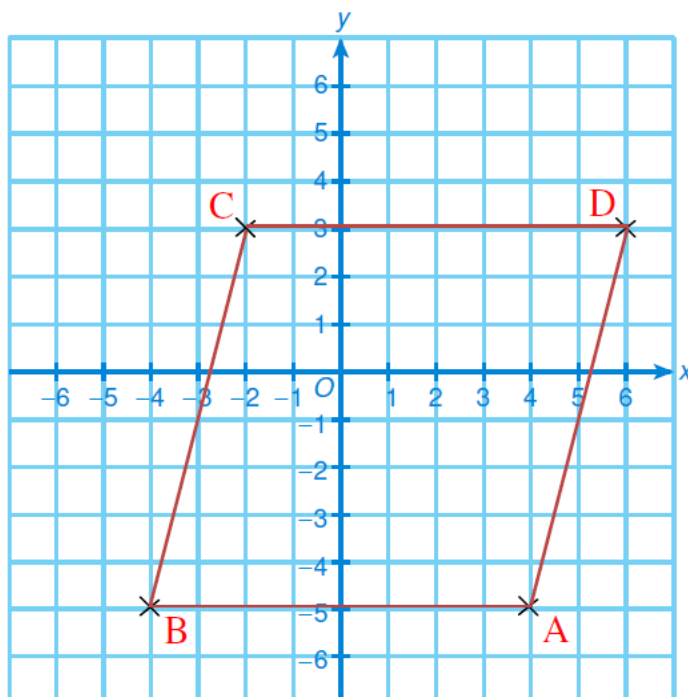
(c) $B = (4, 140^\circ)$ (d) Area = Area of $\triangle OAD \times 4$

$$= \frac{1}{2} \times 4 \times 4 \times \sin 90^\circ$$

$$= 32 \text{ sq units}$$

(1B10E002)

(a)

(b) Coordinates of $D = (6, 3)$

(c) $E = (1, -1)$

It lies in quadrant IV.

(e) Area of $\triangle ABC = [4 - (-4)] \times [3 - (-5)] \div 2$
 $= 32$ sq. units

Area of $\triangle ABE = [4 - (-4)] \times [(-1) - (-5)] \div 2$
 $= 16$ sq. units

Area of $\triangle BCE = 32 - 16$
 $= 16$ sq. units

(1B10E003)

(a) $\therefore \angle POQ = 210^\circ - 120^\circ$
 $= 90^\circ$

1M for considering $\angle POQ$ $\therefore \triangle OPQ$ is a right-angled triangle.

1A with f.t.

(b) The area $= \frac{24 \times 7}{2}$
 $= 84$ sq. units.

1M

1A

(1B10E004)

(a) The coordinates of P and Q are $(3, 40^\circ)$ and $(6, 100^\circ)$ respectively.

(b) $\angle POQ = \angle XOQ - \angle XOP$
 $= 100^\circ - 40^\circ$
 $= \underline{60^\circ}$

1A

(c) In $\triangle POQ$,

$\angle OPQ + \angle POQ + \angle OQP = 180^\circ$

$\angle OPQ + 60^\circ + 30^\circ = 180^\circ$

$\angle OPQ = 180^\circ - 60^\circ - 30^\circ$

$= 90^\circ$

1A

 $\triangle POQ$ is a right-angled triangle. $\therefore$ The claim is agreed.

(1B10E005)

(a) $N = (-3, 0)$

(b) $AC = [6 - (-8)]$
 $= 14$

$BN = 10$

$DN = 6$

$\therefore$ Area of quadrilateral $ABCD = \text{Area of } \triangle ABC - \text{Area of } \triangle ADC$

$= \left(\frac{1}{2} \times 14 \times 10 - \frac{1}{2} \times 14 \times 6\right)$

$= \underline{\underline{28}}$

(1B10E006)

$BC = [1 - (-4)]$ units

1M

$= 5$ units

1A

$\therefore$ Area of square $ABCD = BC \times BC$
 $= 5 \times 5$ units
 $= \underline{\underline{25 \text{ units}}}$

1M

1A

(1B10E007)

- a. $AF = 8 - (-3) = 11$ units [3]
 $2 \times (AB + 11) = 36$
 $AB = 7$ units
- b. $B(-3, -3 + 7) = (-3, 4)$ [2]
 $C(1, 4), E(8, 1)$ [2]
- c. $DE = 8 - 1 = 7$ units [3]
 $CD = 4 - 2 = 2$ units
 Area of $ABCDEF$
 $= 11 \times 7 - 7 \times 2$
 $= 63$ sq. units

(1B10E008)

- (a) $AB = 10 - (-5)$ 1M
 $= 15$ units 1A
- (b)(i) $BC = 60 \div 15$ 1M
 $= 4$ units 1A
- (ii) $C(10, 6)$ 1M for 2 + BC
 $D(-5, 6)$ 1A for both const.
- (c) $(0, 10)$ or $(0, -6)$ 1M for 2 + 8 or 2 - 8
 1A for both const.

Multiple Choice Questions

(1B10F001)

☐ C

(1B10F002)

☐ A

(1B10F003)

☐ B

(1B10F004)

☐ B $\therefore$ Points A and B have the same y -coordinate. $\therefore$ AB is parallel to the x -axis. $\therefore$ Points D and E have the same y -coordinate. $\therefore$ DE is parallel to the x -axis.

(1B10F005)

☐ APoints A and B have the same x -coordinate.The line passes through A and B is parallel to the y -axis.Points A and C have the same y -coordinate.The line passes through A and C is parallel to the x -axis.

(1B10F006)

☐ CThe distance between the origin and $A = \left[0 - \left(-2\frac{5}{11} \right) \right] \text{ units} = \underline{\underline{2\frac{5}{11} \text{ units}}}$

(1B10F007)

☐ B $AB = [-2 - (-4)] \text{ units} = \underline{\underline{2 \text{ units}}}$

(1B10F008)

☐ BIf Q lies on the right of P , then the coordinates of $Q = (-5 + 3, -2) = (-2, -2)$ If Q lies on the left of P , then the coordinates of $Q = (-5 - 3, -2) = (-8, -2)$ If Q lies above P , then the coordinates of $Q = (-5, -2 + 3) = (-5, 1)$ If Q lies below P , then the coordinates of $Q = (-5, -2 - 3) = (-5, -5)$

(1B10F009)

C

- $\therefore AB = [1 - (-4)] \text{ units} = 5 \text{ units}$
 $\therefore$ I is incorrect.
 $\therefore$ Points A and B have the same x -coordinate.
 $\therefore AB$ is parallel to the y -axis.
 $\therefore$ II is correct.
 $\therefore A$ lies in quadrant II and B lies in quadrant III.
 $\therefore AB$ intersects with the x -axis.
 $\therefore$ III is correct.

(1B10F010)

D

- $AB = [x - (x - 1)] \text{ unit} = 1 \text{ unit}$
 $AC = [(y + 2) - y] \text{ units} = 2 \text{ units}$
 $\therefore$ Points A and C have the same x -coordinate.
 $\therefore$ III is correct.

(1B10F011)

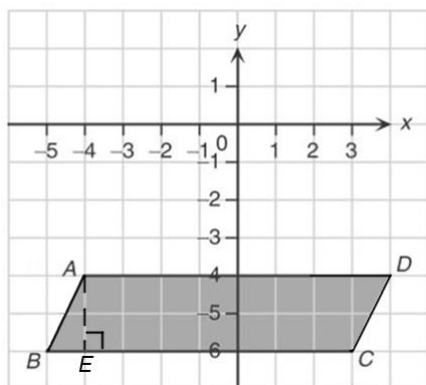
D

$$AB = \left[-1\frac{1}{4} - \left(-3\frac{3}{8} \right) \right] \text{ units} = 2\frac{1}{8} \text{ units}$$

$$\begin{aligned}
 \therefore \text{Perimeter of } \triangle ABC &= 3 \times AB \\
 &= 3 \times 2\frac{1}{8} \text{ units} \\
 &= \underline{\underline{6\frac{3}{8} \text{ units}}}
 \end{aligned}$$

(1B10F012)

C



Construct line segment AE such that $AE \perp BC$.
 Then, the coordinates of $E = (-4, -6)$.

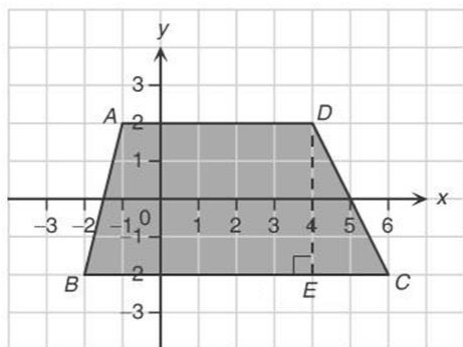
$$BC = [3 - (-5)] \text{ units} = 8 \text{ units}$$

$$AE = [-4 - (-6)] \text{ units} = 2 \text{ units}$$

$$\begin{aligned} \text{Area of parallelogram } ABCD &= BC \times AE \\ &= 8 \times 2 \text{ sq. units} \\ &= \underline{\underline{16 \text{ sq. units}}} \end{aligned}$$

(1B10F013)

C



Construct line segment DE such that $DE \perp BC$.

Then, the coordinates of $E = (4, -2)$.

$$AD = [4 - (-1)] \text{ units} = 5 \text{ units}$$

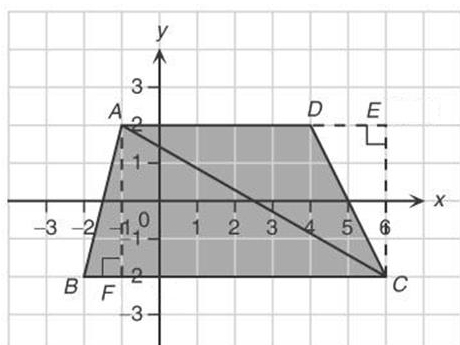
$$BC = [6 - (-2)] \text{ units} = 8 \text{ units}$$

$$DE = [2 - (-2)] \text{ units} = 4 \text{ units}$$

$$\begin{aligned} \text{Area of trapezium } ABCD &= \frac{1}{2} \times (AD + BC) \times DE \\ &= \frac{1}{2} \times (5 + 8) \times 4 \text{ sq. units} \\ &= \underline{\underline{26 \text{ sq. units}}} \end{aligned}$$

(1B10F014)

B



Construct line segment AF such that $AF \perp BC$.

Then, the coordinates of $F = (-1, -2)$.

AF is the common height of $\triangle ABC$ and $\triangle ACD$.

$$BC = [6 - (-2)] \text{ units} = 8 \text{ units}$$

$$AF = [2 - (-2)] \text{ units} = 4 \text{ units}$$

$$AD = [4 - (-1)] \text{ units} = 5 \text{ units}$$

$$\begin{aligned} \text{Area of } \triangle ABC &= \frac{1}{2} \times BC \times AF \\ &= \frac{1}{2} \times 8 \times 4 \text{ sq. units} \\ &= 16 \text{ sq. units} \end{aligned}$$

$$\begin{aligned} \text{Area of } \triangle ACD &= \frac{1}{2} \times AD \times AF \\ &= \frac{1}{2} \times 5 \times 4 \text{ sq. units} \\ &= 10 \text{ sq. units} \end{aligned}$$

$$\begin{aligned} \therefore \text{Difference} &= \text{area of } \triangle ABC - \text{area of } \triangle ACD \\ &= (16 - 10) \text{ sq. units} \\ &= \underline{\underline{6 \text{ sq. units}}} \end{aligned}$$

(1B10F015)

B

$$\text{Base of } \triangle PQR = QR = [4 - (-6)] \text{ units} = 10 \text{ units}$$

$$\text{Height of } \triangle PQR = [-2 - (-6)] \text{ units} = 4 \text{ units}$$

$$\begin{aligned} \therefore \text{Area of } \triangle PQR &= \frac{1}{2} \times 10 \times 4 \text{ sq. units} \\ &= \underline{\underline{20 \text{ sq. units}}} \end{aligned}$$

(1B10F016)

A

$$\begin{aligned} \text{The coordinates of } A_1 &= (-2 + 3, 1 + 2) \\ &= \underline{\underline{(1, 3)}} \end{aligned}$$

(1B10F017)

C

$$\begin{aligned} \text{The coordinates of } R' &= (-3, -(-2)) \\ &= \underline{\underline{(-3, 2)}} \end{aligned}$$

(1B10F018)

C

The distance between C and L $=$ the distance between C' and L $= (6 - 4)$ units $= 2$ unitsThe coordinates of C' $= (5, 6 - 2 \times 2)$ $= \underline{\underline{(5, 2)}}$

(1B10F019)

A

The distance between D and L $=$ the distance between D' and L $= [1 - (-2)]$ units $= 3$ unitsThe coordinates of D' $= (1 - 2 \times 3, -1)$ $= \underline{\underline{(-5, -1)}}$

(1B10F020)

A

 L must pass through the points with y -coordinate $\frac{-8 - (-1)}{2} = -3.5$ $\therefore L$ must pass through $(5, -3.5)$.

(1B10F021)

B

The coordinates of $A' = (-3, -(-9)) = \underline{\underline{(-3, 9)}}$

(1B10F022)

D

The coordinates of $T' = \underline{\underline{(-b, -a)}}$

(1B10F023)

D

 $U = (-2, 8)$ $W = \underline{\underline{(8, 2)}}$

(1B10F024)

B

$$P = \underline{\underline{(2, -3)}}$$

$$Q = \underline{\underline{(3, 2)}}$$

(1B10F025)

B

When P is reflected about the y -axis, the image is $(-a, b)$.

B is correct.

(1B10F026)

D

$\therefore P$ is rotated anti-clockwise about the origin O through 180° to Q .

$$\therefore OP = OQ$$

$\therefore$ I is true.

$$\therefore a = -c, b = -d$$

$\therefore$ II is true but III is not true.

$\therefore$ The answer is D.

(1B10F027)

D

K is the image of J when J is rotated anti-clockwise about O through 90° .

$$OJ = OK$$

I is correct.

The coordinates of $K = (-y, x)$

II is correct.

From the figure, the coordinates of $M = (x, 0)$

III is correct.

$\therefore$ The answer is D.

(1B10F028)

D

(1B10F029)

C

$$\begin{aligned}\angle COD &= 150^\circ - 30^\circ \\ &= \underline{\underline{120^\circ}}\end{aligned}$$

(1B10F030)

B

$$\therefore 304^\circ - 124^\circ = 180^\circ$$

1B Chapter 10 Quiz

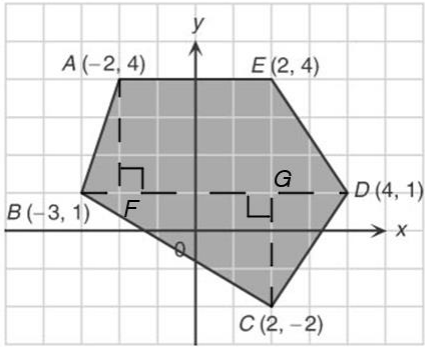
Introduction to Coordinates

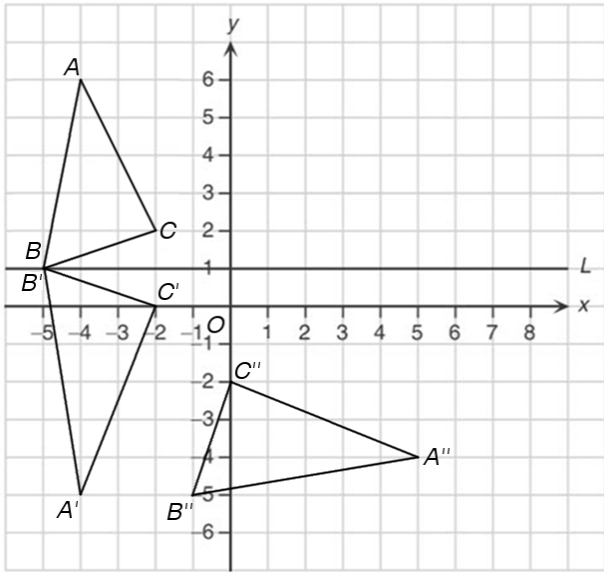
Solutions

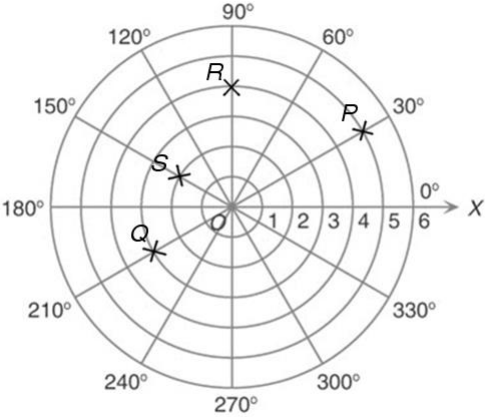
* Deduct at most 1 mark for poor presentation (pp – 1) in the whole quiz.

** Deduct at most 1 mark for wrong units or no units (u – 1) in the whole quiz.

Question	Answers/Solutions	Marks	Remarks
1	$A = (-4, 1)$, $B = (0, -2)$	0.5 0.5	
2		1	0.5A for plotting C 0.5A for plotting D
3	A lies in quadrant II, D lies in quadrant IV.	0.5 0.5	
4	B lies on the y-axis, C lies on the x-axis.	0.5 0.5	
5	7 units	1	u – 1 for no units
6	2 units	1	u – 1 for no units
7	68	1	
8	17	1	
9	(a) (0, 2) (b) (3, 2)	0.5 0.5	

Question	Answers/Solutions	Marks	Remarks
10	(a) $(-6, -1)$ (b) $(5, 4)$	0.5 0.5	
11	(a) Length of the path $ABCDE$ $= AB + BC + CD + DE$ $= (AB + CD) + (BC + DE)$ $= \{[5 - (-5)] + [5 - (-5)]\} \text{ units}$ $= \underline{20 \text{ units}}$ (b) Length of the path $AFGHIJE$ $= AF + FG + GH + HI + IJ + JE$ $= FG + HI + JE + AF + GH + IJ$ $= \{[5 - (-5)] + [4 - (-5)] + (4 - 2) + (5 - 2)\} \text{ units}$ $= \underline{24 \text{ units}}$	1 1 1	Method mark u – 1 for no units Method mark u – 1 for no units
12	(a)  (b) Take BD as the common base of $\triangle BCD$ and trapezium $ABDE$. Draw the height CG of $\triangle BCD$ and the height AF of trapezium $ABDE$. Then, the coordinates of F and G are $(-2, 1)$ and $(2, 1)$ respectively. $AE = [2 - (-2)] \text{ units} = 4 \text{ units}$ $BD = [4 - (-3)] \text{ units} = 7 \text{ units}$ $AF = (4 - 1) \text{ units} = 3 \text{ units}$ $CG = [1 - (-2)] \text{ units} = 3 \text{ units}$ Area of pentagon $ABCDE$	1	pp – 1 for poor presentation. u – 1 for no units.

Question	Answers/Solutions	Marks	Remarks
	$= \frac{1}{2} \times (AE + BD) \times AF + \frac{1}{2} \times BD \times CG$ $= \frac{1}{2} \times (4 + 7) \times 3 + \frac{1}{2} \times 7 \times 3 \text{ sq. units}$ $= \underline{\underline{27 \text{ sq. units}}}$	1 1 1	
13	(a) The coordinates of $A' = (2 + 4, -5)$ $= \underline{\underline{(6, -5)}}$ (b) The coordinates of $B' = (-(-2), -3)$ $= \underline{\underline{(2, -3)}}$ (c) The coordinates of $C' = \underline{\underline{(9, -4)}}$ (d) The coordinates of $D' = \underline{\underline{(-1, -5)}}$	1 1 1 1	
14	 (a) $A'(-4, -4)$, $B'(-5, 1)$, $C'(-2, 2)$ (b) $A''(4, -4)$, $B''(-1, -5)$, $C''(0, -2)$	2 1 1	1M for drawing each triangle

Question	Answers/Solutions	Marks	Remarks
15	(a)  (b) $PQ = 5 + 3 = \underline{8 \text{ units}}$ (c) $\angle ROS = 150^\circ - 90^\circ = \underline{60^\circ}$	2	0.5A for plotting a correct point
		1	u – 1 for no units
		1	u – 1 for no units